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SYSTEM ANALYSIS

2.1 Introduction

The analysis phase is one of the most important stages in the software and embedded systems development life cycle because it defines the operational expectations, user needs, technical requirements, and system behavior before implementation begins. During this phase, the proposed Smart Energy Management Adapter with Per-Equipment Monitoring is examined from both functional and operational perspectives in order to determine how the system should operate, interact with users, process data, and respond to real-world conditions.

The purpose of this chapter is to provide a detailed understanding of the problem domain, identify the requirements of the proposed system, analyze user interactions, and evaluate the feasibility of implementing the solution. This phase establishes a strong foundation for system design, implementation, testing, and deployment.

The proposed system is intended to address challenges associated with inefficient energy utilization, lack of appliance-level monitoring, delayed fault detection, and limited user visibility into electrical consumption patterns. By integrating embedded systems, wireless communication, cloud technologies, and intelligent monitoring techniques, the system provides users with a centralized platform for monitoring and controlling electrical appliances in real time.

The analysis chapter therefore focuses on:

- Functional requirements of the system
- Non-functional requirements
- User interactions and operational expectations
- Communication and processing requirements
- Feasibility analysis
- System constraints and assumptions
- Data flow requirements

The findings from this analysis guide the architecture and implementation presented in the design chapter.

2.1.1 Problem Statement

In many homes, offices, hostels, and small business environments, electrical energy consumption is monitored only at the total utility meter level. This makes it difficult for users to identify which specific appliances consume the most energy or contribute to excessive electricity bills.

Additionally, traditional electrical systems lack:

- Real-time appliance-level monitoring
- Remote switching capabilities
- Intelligent overload management
- Automated energy optimization
- Historical energy analytics
- Instant fault notification systems

As a result, users experience:

- Increased electricity costs
- Energy wastage
- Delayed response to abnormal conditions
- Difficulty managing multiple appliances
- Limited visibility into energy usage behavior

The proposed Smart Energy Management Adapter addresses these limitations by providing a low-cost, intelligent, and scalable platform capable of monitoring, analyzing, and controlling electrical appliances individually.

2.1.2 Objectives of the Proposed System

General Objective

To design and develop a Smart Energy Management Adapter capable of monitoring, analyzing, and controlling electrical appliances at individual equipment level using embedded systems and wireless communication technologies.

Specific Objectives

The specific objectives of the project are:

- To measure voltage and current consumption of multiple electrical appliances
 - To calculate real-time power and accumulated energy usage
 - To provide remote monitoring through mobile and web interfaces
 - To enable remote appliance switching using relay control
 - To implement overload protection and threshold-based automation
 - To support cloud-based data storage and analytics
 - To generate alerts and notifications during abnormal operation
 - To simulate the entire system in software due to hardware limitations
-

2.1.3 Scope of the System

The proposed system focuses on the monitoring and management of electrical appliances within indoor environments such as:

- Residential homes
- Student hostels
- Offices
- Laboratories
- Small business premises

The system monitors:

- Appliance current consumption
- Supply voltage
- Instantaneous power
- Accumulated energy usage
- Appliance operational status

The project includes:

- Embedded system simulation
- Wireless communication simulation
- Cloud integration concepts
- User dashboard interfaces
- Appliance control logic
- Data analytics and reporting

The project does not include:

- Industrial three-phase energy management
 - Utility-grade billing certification
 - High-voltage industrial automation
 - Renewable energy integration
-

2.1.4 Assumptions and Constraints

Assumptions

The system design assumes that:

- Stable AC mains supply is available
- Users possess internet-enabled devices
- WiFi connectivity exists within the operational area
- Sensors are calibrated correctly
- Appliances operate within expected current ranges

Constraints

The project faces several constraints including:

- Limited access to physical hardware components
- Budget limitations for prototype development
- Dependence on wireless network availability
- Processing limitations of embedded microcontrollers
- Time limitations associated with academic projects

To overcome hardware limitations, the system is implemented and tested using software simulation techniques.

2.1.5 Existing System Analysis

Traditional energy monitoring systems primarily measure total electrical consumption without providing detailed appliance-level visibility. Most conventional systems also lack intelligent automation and remote control functionality.

Limitations of Existing Systems

Existing systems commonly suffer from:

- Lack of per-appliance monitoring
- Manual switching operations
- Absence of cloud connectivity
- Poor scalability
- Delayed fault detection
- Limited data visualization capabilities
- No predictive or analytical functionality

Advantages of the Proposed System

The proposed system improves upon traditional approaches by providing:

- Real-time appliance-level monitoring
- Wireless remote access
- Intelligent relay control

- Historical analytics
 - Automated alerts and notifications
 - Cloud synchronization
 - Expandable modular architecture
-

2.1 Introduction

This chapter presents the system analysis for the proposed Smart Energy Management Adapter with Per-Equipment Monitoring. The analysis phase identifies the functional and non-functional requirements of the system, the expected system behavior, user interactions, operational constraints, and feasibility considerations.

The purpose of this phase is to clearly define what the system should accomplish before implementation and testing.

2.2 Functional Requirements

Functional requirements describe the specific operations and services the system must perform.

2.2.1 Real-Time Energy Monitoring

The system shall:

- Measure voltage and current continuously
 - Calculate instantaneous power consumption
 - Compute accumulated energy usage (kWh)
 - Monitor individual appliances independently
 - Display real-time consumption values
-

2.2.2 Appliance Control

The system shall:

- Allow remote ON/OFF control of appliances
 - Support relay-based switching
 - Enable scheduled switching operations
 - Disconnect appliances during overload conditions
 - Restore appliances automatically when safe
-

2.2.3 User Authentication and Access

The system shall:

- Support secure user login
 - Restrict unauthorized access
 - Maintain user profiles and permissions
 - Authenticate cloud communication requests
-

2.2.4 Wireless Communication

The system shall:

- Support WiFi communication
 - Support Bluetooth Low Energy (BLE)
 - Transmit data to cloud services
 - Receive remote user commands
 - Maintain communication reliability
-

2.2.5 Alerts and Notifications

The system shall:

- Detect overload conditions
 - Generate fault notifications
 - Send high-consumption alerts
 - Notify users during abnormal operation
 - Support SMS/Telegram notifications
-

2.2.6 Data Logging and Storage

The system shall:

- Store historical energy consumption data
 - Maintain appliance usage records
 - Support cloud-based storage
 - Enable export of reports in CSV/PDF format
 - Preserve timestamps for all measurements
-

2.2.7 Dashboard and Visualization

The system shall:

- Display charts and graphs
 - Show per-appliance statistics
 - Display total power usage
 - Present historical trends
 - Provide mobile and web interfaces
-

2.2.8 System Automation

The system shall:

- Execute threshold-based control logic
 - Support automatic relay switching
 - Implement load shedding during overload
 - Support configurable operational rules
-

2.3 Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints of the system.

2.3.1 Performance Requirements

The system should:

- Respond to user commands within a few seconds
 - Process sensor data in real time
 - Support continuous monitoring operation
 - Maintain low latency communication
-

2.3.2 Reliability Requirements

The system should:

- Operate continuously with minimal downtime
 - Maintain accurate sensor readings
 - Recover automatically after communication failures
 - Continue local operation without internet connectivity
-

2.3.3 Scalability Requirements

The system should:

- Support multiple appliances
 - Allow future sensor expansion
 - Support integration with additional cloud services
 - Enable future smart automation features
-

2.3.4 Security Requirements

The system should:

- Encrypt sensitive communication
 - Protect user credentials
 - Prevent unauthorized access
 - Secure cloud transactions
-

2.3.5 Usability Requirements

The system should:

- Provide a simple user interface
 - Display clear monitoring information
 - Support easy device configuration
 - Be accessible from mobile and web platforms
-

2.4 System Users

The primary users of the system include:

- Homeowners

- Hostel managers
- Small business operators
- Facility administrators
- Energy management personnel

These users interact with the system to monitor and manage electrical energy consumption.

2.5 Feasibility Analysis

2.5.1 Technical Feasibility

The project is technically feasible because:

- ESP32 microcontrollers support WiFi and Bluetooth
 - Current and voltage sensors are readily available
 - Cloud platforms support IoT integration
 - Software simulation can replace unavailable hardware during development
-

2.5.2 Economic Feasibility

The proposed system is economically feasible because:

- Components are relatively low-cost
 - The system reduces energy wastage
 - Open-source software tools minimize development cost
 - Simulation reduces hardware prototyping expenses
-

2.5.3 Operational Feasibility

The system is operationally feasible because:

- Users can easily operate the interfaces
 - Remote monitoring improves convenience
 - Automated control reduces manual intervention
 - The architecture supports real-time operation
-

2.6 Summary

This chapter presented the system analysis for the Smart Energy Management Adapter with Per-Equipment Monitoring.

The analysis identified the functional requirements, non-functional requirements, users, and feasibility of the proposed system.

These requirements form the foundation for the system design and implementation presented in the next chapter.

SYSTEM DESIGN

3.1 Introduction

This chapter presents the design and architecture of the proposed Smart Energy Management Adapter with Per-Equipment Monitoring. The system is designed to monitor, analyze, and control electrical energy consumption at appliance level using embedded systems, wireless communication, cloud integration, and intelligent control logic.

The architecture follows a layered approach that separates user interaction, communication, processing, sensing, actuation, and power management into modular components. This modular organization improves scalability, maintainability, reliability, and ease of future expansion.

3.2 System Architecture

3.2.1 System Architecture Diagram

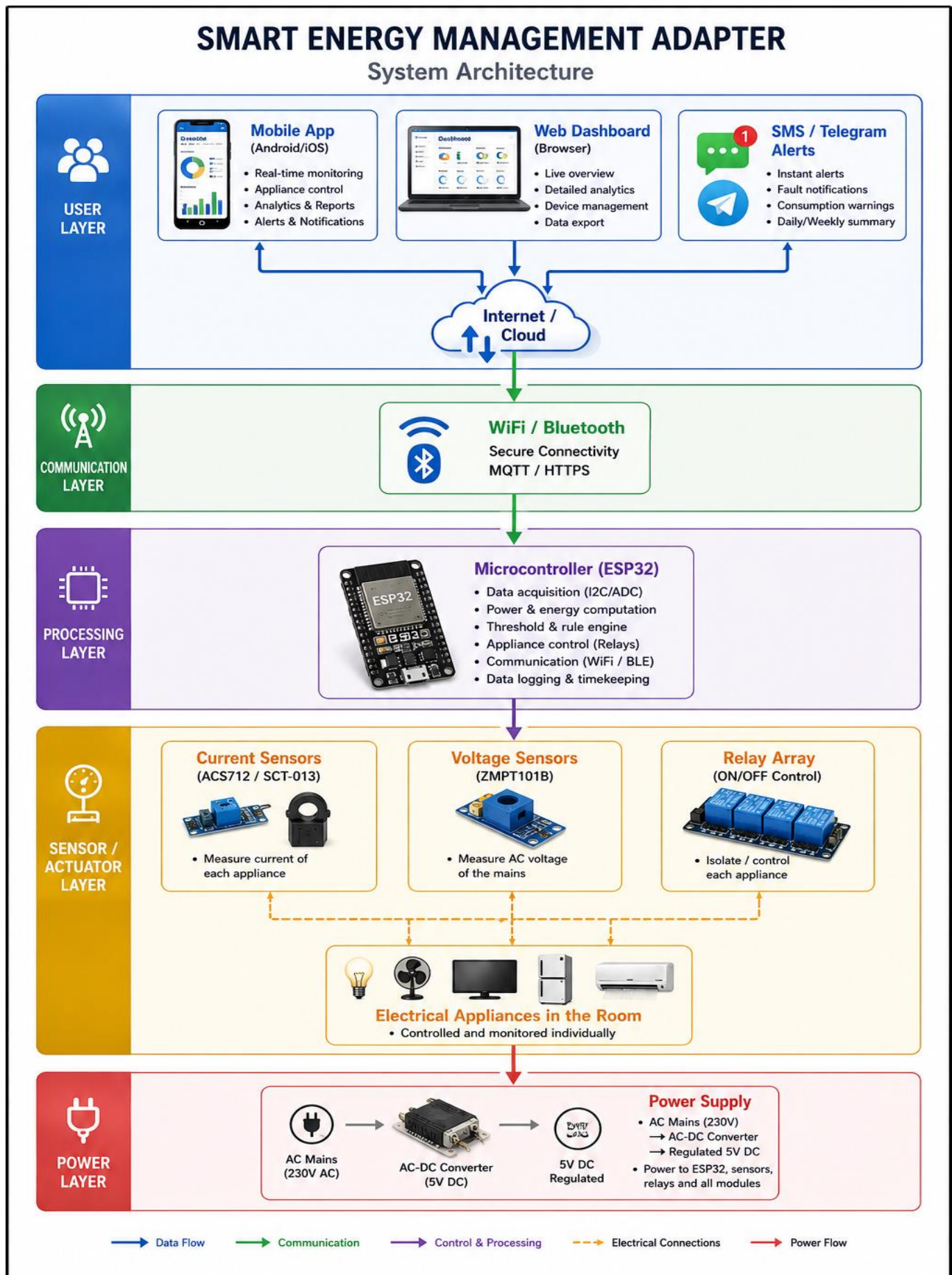


Figure 3.1: Smart Energy Management System Architecture

The system architecture consists of five major layers:

1. User Layer
2. Communication Layer
3. Processing Layer
4. Sensor/Actuator Layer
5. Power Layer

These layers work together to provide real-time energy monitoring, appliance control, data analysis, and remote access functionality.

3.3 Explanation of System Architecture Layers

3.3.1 User Layer

The User Layer is the topmost layer of the architecture and serves as the interaction point between the user and the system.

This layer consists of:

- Mobile Application (Android/iOS)
- Web Dashboard
- SMS/Telegram Notification System

Functions of the User Layer

The user layer enables users to:

- Monitor real-time power consumption
- View appliance-specific energy usage
- Turn appliances ON/OFF remotely
- Configure power thresholds
- Receive alerts and notifications
- View historical analytics and reports
- Export energy reports in CSV/PDF format

Mobile Application

The mobile application provides portable access to the system. It allows users to:

- View live appliance status
- Track total energy consumption
- Receive instant notifications
- Configure schedules and thresholds
- Access historical consumption trends

Web Dashboard

The web dashboard provides a more detailed monitoring platform with:

- Advanced analytics
- Multi-room monitoring
- Graphical reports
- Administrative settings
- Device management

SMS/Telegram Alerts

This subsystem ensures users without smartphones or internet access can still receive:

- Fault alerts
 - Overload warnings
 - Consumption notifications
 - Scheduled reports
-

3.3.2 Communication Layer

The Communication Layer manages all wireless data transmission between hardware devices and user interfaces.

Communication Technologies

The system uses:

- WiFi Communication
- Bluetooth Low Energy (BLE)
- MQTT Protocol
- HTTP/HTTPS Protocols

Functions of the Communication Layer

This layer is responsible for:

- Sending sensor data to cloud services
- Receiving user commands
- Managing network connections
- Encrypting transmitted data
- Handling reconnection and retransmission
- Device discovery and configuration

WiFi Communication

WiFi provides internet connectivity for:

- Cloud synchronization
- Remote monitoring
- Mobile application communication
- Dashboard access
- Firmware updates

Bluetooth Communication

Bluetooth Low Energy (BLE) is used for:

- Initial device setup
- Local configuration
- Offline monitoring
- Device pairing

Communication Flow

The Communication Layer performs bidirectional communication:

Upstream Communication

Sensor Layer → Processing Layer → Communication Layer → Cloud/User Interfaces

Downstream Communication

User Interfaces → Cloud → Communication Layer → Processing Layer → Relay Control

3.3.3 Processing Layer

The Processing Layer forms the intelligence core of the system.

The ESP32 microcontroller is used because of its:

- Integrated WiFi and Bluetooth
- Dual-core processing capability
- Multiple GPIO pins
- ADC and I2C support
- Low power consumption
- Real-time processing capability

Functions of the Processing Layer

The ESP32 performs:

- Sensor data acquisition
- Signal calibration
- Power calculations
- Energy accumulation
- Threshold comparison
- Load management
- Relay control
- Communication handling
- Fault detection
- Timekeeping and scheduling

Data Acquisition

The ESP32 continuously reads:

- Current sensor outputs
- Voltage sensor outputs
- ADC converted values

Power Calculation

Electrical power is calculated using:

$$P = V \times I$$

Where:

- P = Power (Watts)
- V = Voltage (Volts)
- I = Current (Amperes)

Energy Calculation

Energy consumption is accumulated over time using:

$$E = P \times t$$

Where:

- E = Energy (kWh)
- P = Power (kW)
- t = Time (hours)

Edge Processing

The Processing Layer performs local processing to ensure:

- Fast response time
- Offline functionality
- Reduced cloud dependency
- Reduced bandwidth usage
- Immediate protection actions

Example Calculation

If:

- ADC reading = 16384
- Sensor sensitivity = 185 mV/A
- Voltage = 230V

Then:

$$I = (16384 \times 5 \times 185) / (65535 \times 1000)$$

$$I \approx 0.23A$$

Power:

$$P = 230 \times 0.23$$

$$P \approx 52.9W$$

Energy after 1 hour:

$$E = 52.9 / 1000$$

$$E \approx 0.0529 \text{ kWh}$$

3.3.4 Sensor and Actuator Layer

The Sensor/Actuator Layer interfaces directly with the electrical environment.

This layer contains:

- Current Sensors
- Voltage Sensors
- Relay Array
- Connected Electrical Appliances

Current Sensors

Current sensors monitor the current consumption of individual appliances.

Sensor Types

The design supports:

- ACS712 Current Sensor
- SCT-013 Current Transformer

Functions

Current sensors:

- Detect appliance current draw
- Enable per-appliance monitoring
- Support overload detection
- Enable energy calculation

Output

The sensors produce analog voltage signals proportional to measured current.

Voltage Sensors

The voltage sensing subsystem measures the mains voltage.

Sensor Used

- ZMPT101B Voltage Sensor Module

Functions

Voltage sensors:

- Measure AC mains voltage
 - Enable accurate power computation
 - Detect abnormal voltage conditions
 - Support calibration and correction
-

Relay Array

The relay array provides appliance switching functionality.

Functions

Relays are used to:

- Turn appliances ON/OFF
- Disconnect overloaded devices
- Perform scheduled switching
- Execute automation logic

Electrical Isolation

Relays provide electrical isolation between:

- Low-voltage microcontroller circuitry
- High-voltage AC appliance circuits

Relay Control Logic

- HIGH signal → Relay ON
 - LOW signal → Relay OFF
-

Connected Appliances

The system supports monitoring and control of multiple electrical appliances such as:

- Fans
- Televisions
- Refrigerators
- Heaters
- Water pumps
- Lighting systems
- Air conditioners

Each appliance can be monitored independently.

3.3.5 Power Layer

The Power Layer provides stable regulated DC power to all electronic components.

Power Source

The system uses:

- 230V AC mains input

Power Conversion

The AC supply is converted into regulated 5V DC using:

- AC-DC Converter
- Voltage Regulator Module

Functions of the Power Layer

The power subsystem:

- Powers the ESP32
- Supplies current sensors
- Powers relays and peripherals
- Protects circuits from fluctuations
- Ensures stable operation

Importance of Regulation

Stable voltage regulation prevents:

- Sensor inaccuracies
- Communication instability
- Microcontroller resets
- Relay malfunction



3.4 System Block Diagram

3.4.1 Hardware and Software Block Diagram

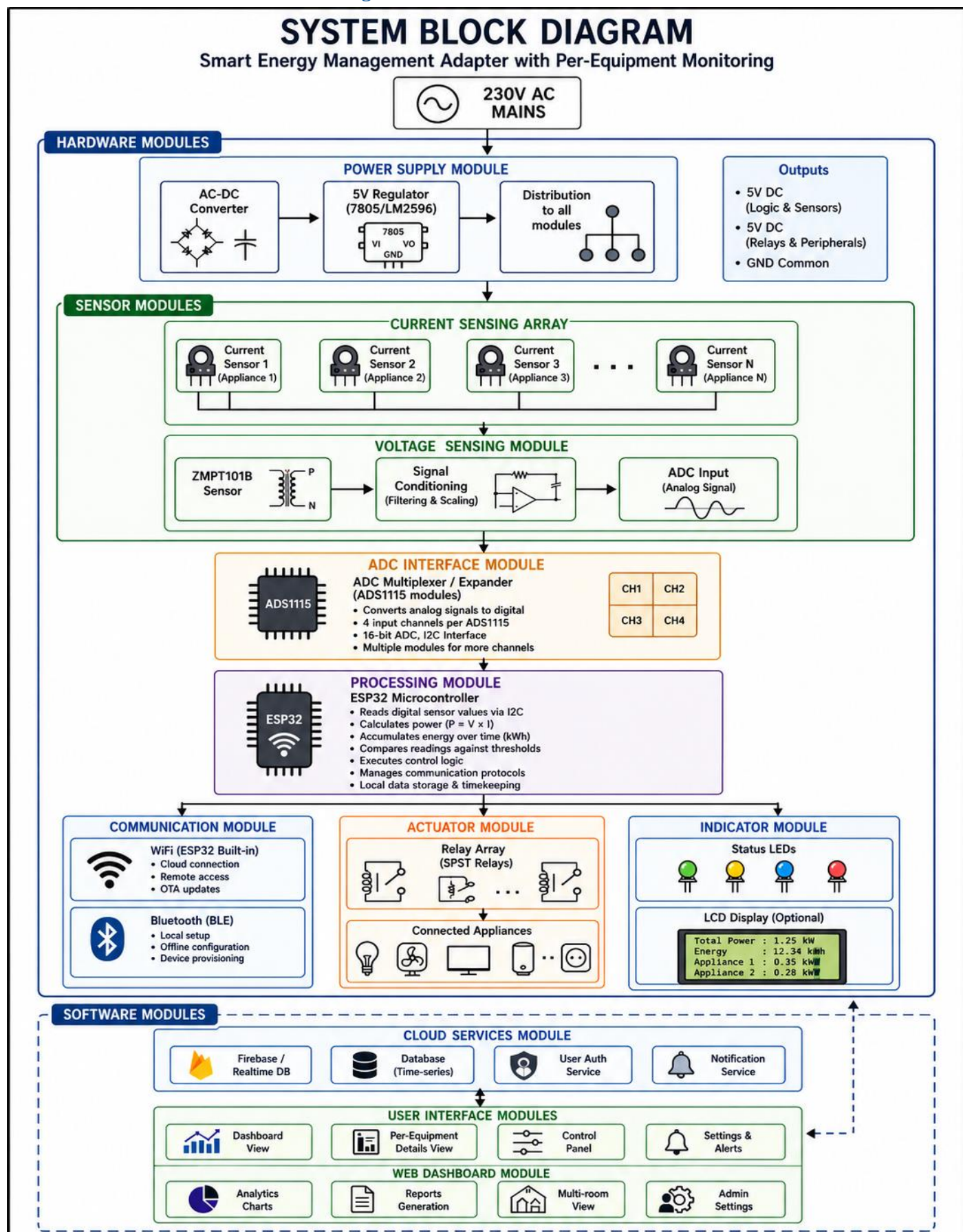


Figure 3.2: Smart Energy Management System Block Diagram

The block diagram illustrates the internal organization of hardware and software modules used in the system.

The system is divided into:

- Hardware Modules
 - Software Modules
-

3.5 Hardware Modules

3.5.1 Power Supply Module

The Power Supply Module converts 230V AC mains into regulated DC voltage required by electronic components.

Components

- AC-DC Converter
- 5V Voltage Regulator
- Distribution Network

Outputs

- 5V DC for logic circuits
 - 5V DC for relays and sensors
 - Common system ground
-

3.5.2 Sensor Modules

The Sensor Module consists of:

- Current Sensing Array
- Voltage Sensing Module

Current Sensing Array

The array supports multiple current sensors for monitoring individual appliances.

Each sensor:

- Monitors a dedicated appliance
- Produces analog current data
- Sends data to ADC modules

Voltage Sensing Module

The voltage sensing subsystem:

- Measures AC mains voltage
 - Conditions analog signals
 - Sends calibrated data to ADC inputs
-

3.5.3 ADC Interface Module

The ADC Interface Module converts analog sensor signals into digital values.

ADC Used

- ADS1115 ADC Module

Functions

The ADC:

- Converts analog voltage/current signals
- Provides multiple input channels
- Interfaces with ESP32 via I2C
- Improves measurement resolution

Advantages

- 16-bit resolution
 - Multiple channels
 - Accurate measurements
 - Noise reduction capability
-

3.5.4 Processing Module

The Processing Module is implemented using an ESP32 microcontroller.

Responsibilities

The module:

- Reads sensor values
- Performs power calculations
- Accumulates energy consumption
- Executes automation logic
- Controls relays
- Manages communication
- Generates alerts

Processing Tasks

- Threshold comparison
 - Overload protection
 - Data formatting
 - Local storage
 - Scheduling
-

3.5.5 Communication Module

The Communication Module handles wireless communication.

Communication Technologies

- WiFi
- Bluetooth Low Energy (BLE)

Functions

The module:

- Connects to cloud services
- Enables remote access
- Supports mobile applications

- Handles local setup and provisioning

Data Flow Example

The Communication Module receives:

- Sensor data from Processing Module
- WiFi credentials
- User commands

It processes:

- JSON formatting
- MQTT packet creation
- Encryption
- Connection management

It outputs:

- Cloud packets
- Device status information
- Parsed commands

Example JSON Payload

```
{
  "device_id": "SMA_001",
  "timestamp": "2026-03-14T10:30:00Z",
  "appliances": [
    {
      "id": 1,
      "name": "Refrigerator",
      "power_w": 150,
      "energy_kwh": 1.2,
      "status": "on"
    },
    {
      "id": 2,
      "name": "TV",
      "power_w": 80,
      "energy_kwh": 0.4,
      "status": "off"
    },
    {
      "id": 3,
      "name": "Water Pump",
      "power_w": 750,
      "energy_kwh": 2.5,
      "status": "on"
    }
  ],
  "total_power_w": 980,
  "alerts": []
}
```

3.5.6 Actuator Module

The Actuator Module consists of relay arrays.

Inputs

- Digital control signals from ESP32

Processes

- Signal amplification
- Coil energization
- Electrical isolation

Outputs

- Physical switching of appliance power
-

3.5.7 Indicator Module

The Indicator Module provides visual feedback.

Components

- Status LEDs
- LCD Display (Optional)

Functions

The module:

- Displays system state
 - Shows appliance consumption
 - Indicates faults and alerts
 - Displays operational status
-

3.6 Software Modules

3.6.1 Cloud Services Module

Cloud services provide remote data storage and analytics.

Functions

Cloud services:

- Store time-series energy data
- Manage user authentication
- Generate notifications
- Support historical analysis
- Synchronize devices

Outputs

- Historical analytics
 - Notifications
 - Authentication tokens
 - Device commands
-

3.6.2 User Interface Modules

The user interface modules provide visualization and control.

Features

The interfaces support:

- Real-time monitoring
- Graphical analytics
- Appliance control
- Alert notifications
- Historical reporting
- Exportable reports

Data Visualization

Data is displayed using:

- Charts
- Graphs
- Numerical displays
- Dashboard widgets
- Status indicators

3.7 Summary of System Data Flow

The following stages summarize how data flows through the system:

Stage	Data Type	Format	Destination
1	Raw analog signals	Voltage (0–5V)	ADC Module
2	Digital sensor values	16-bit integers	Processing Module
3	Calibrated measurements	Floating point values	Internal Processing
4	Processed power data	JSON strings	Communication Module
5	Formatted packets	MQTT/HTTP JSON	Cloud Services
6	Visualized data	Charts/Graphs	User Interfaces
7	User commands	MQTT/HTTP JSON	Processing Module
8	Control signals	HIGH/LOW digital	Relay Array

3.8 Summary

This chapter presented the complete system design and architecture of the Smart Energy Management Adapter with Per-Equipment Monitoring.

The proposed architecture integrates:

- Embedded systems
- Wireless communication
- Cloud computing
- Real-time monitoring
- Appliance control

- Data analytics

The layered architecture improves modularity, scalability, and maintainability while supporting future enhancements such as machine learning, predictive analytics, and smart automation.

The next chapter presents the implementation methodology and simulation of the proposed system.